

Pilot Studies for the Second Dissertation Paper: Design, Full Test Battery, and Results

Version 2 — including the definitive rerun with exact nominal legs

Pilot I: Regime-Consistent Resampling (simulation) — Pilot II: The TIPS–Treasury Basis
Floor (Bloomberg)

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Abstract

This note documents, at full technical detail, the feasibility pilots executed for the candidate second dissertation papers: every test run, every number obtained, and every verdict against the pre-registered criteria. Pilot I (regime-consistent resampling; zero market data) shows that naive resampling of rate increments produces a ten-fold violation rate on a nominal 1% VaR and more than doubles the size of a nominal 5% predictability test under the published three-regime volatility calibration, while the “disguise” transformation restores both to near-nominal: the project’s premise is demonstrated. Pilot II (the TIPS–Treasury floor) was run twice: a first pass on the original Bloomberg extract, and a definitive rerun after four exact nominal constant-maturity series, five breakeven series, swap spreads and the NSA CPI were added. The definitive results: the basis proxy averages 46.8 bp over the exact Fleckenstein–Longstaff–Lustig window (their bond-level figure: 54.5 bp), validating the construction, which is further certified by a breakeven cross-check (10-year: mean difference -0.55 bp, correlation 0.996); the post-2013 floor is 16–24 bp at 5–30 years with Newey–West t -statistics up to 24; the compression is a *staircase*, with the dominant break at December 2010 ($\text{sup-}F = 160$) and a second at December 2017 — the first pass’s “2020–21 second wave” is exposed as a sample-truncation artefact; the apparent term-structure inversion of the first pass dissolves once legs are exact, into a rising 5–10 segment plus two identified artefacts (a 52 bp seasonal amplitude at 2 years, formally significant, and the cheap-20-year-sector effect); the 10-year basis is stationary with a 4.6-month half-life; and stress episodes move the basis three-to-seven-fold while *unconditional linear* loadings on MOVE are indistinguishable from zero — a pattern that rejects a linear-factor description and points exactly to the state-dependent, barrier-type mechanism the structural model prices. Net verdict: GO, with the empirical design steered toward stress-conditional specifications and bond-level construction at the short end and for March 2020.

1 Purpose, Scope, and the Go/No-Go Logic

The three leading candidates for the second dissertation paper — the structural TIPS–Treasury floor model, the regime-consistent resampling paper, and the convexity-premium paper — were each designed with an explicit feasibility gate: a cheap, decisive test of the project’s empirical premise, executed *before* committing research time, with pre-registered pass/fail criteria. This note reports the two gates executed so far. Pilot I requires no market data. Pilot II required a small Bloomberg extract (TIPS_variables.xlsx) and, for the definitive rerun, a supplementary extract (Cartel1.xlsx: USGG2YR/5YR/10YR/30YR, USGGBE02/05/10/20/30, USSP2/5/10/30, CPURNSA). The convexity pilot, whose binding risk admits no data shortcut, remains scheduled as its own two-week Phase-0 exercise. The reporting convention is conservative throughout: shortcuts are flagged, failed criteria are stated as failed, and first-pass findings later overturned by better data are reported as overturned, with the mechanism of the error identified.

2 Pilot I — Regime-Consistent Resampling: the Zero-Data Kill Test

2.1 Question and design

The resampling paper’s premise is that naive bootstrap schemes — which glue historical rate increments onto synthetic paths regardless of the level at which those increments occurred — produce materially distorted inference when the data-generating process has the three-regime, level-dependent volatility of De Guillaume–Rebonato–Pogudin (2013) and Rebonato–El Aouadi (2021), and that resampling in the “disguised” space (the variance-stabilising transformation Σ) repairs the distortion. The pilot builds the laboratory maximally favourable to the mechanism: DGP $dr_t = s(r_t) dz_t$ with the modified three-segment sigma function and the published combined calibration ($y_L = 2.65\%$, $y_R = 7.05\%$, $K = 27.8$, $\alpha = 1.03\%$; $\sigma_G = 25$ bp/month); disguise and inverse implemented in closed form with continuity of level and slope at the switch points, verified to machine precision; paths clipped to $(-\alpha, 20\%)$; fixed seed (42); full reproducibility from `pilot_resampling.py`.

2.2 Exhibit A — Value-at-Risk coverage under regime mismatch

A 600-month history confined to the normal regime (the analogue of a risk system estimated on the 3–6% era) feeds one-month historical-simulation VaR, applied at current levels r_{now} ; true coverage is measured against 200,000 draws of the true move, whose volatility is $s(r_{\text{now}})$. The naive scheme fails by an order of magnitude in both directions (10.1% breaches on a 1% VaR at high rates; zero breaches, i.e. wasted capital, at low rates); the disguise restores near-nominal coverage at every level. Two internal checks: at $r_{\text{now}} = 4\%$ — inside the sample’s regime — the schemes coincide exactly, confirming regime mismatch as the sole mechanism; and the disguised residual above nominal (1.6% vs 1%) is the finite-sample-quantile plus back-transformation bias that the paper’s theory layer is designed to correct.

Nominal	r_{now}	Naive HS breach	Disguised breach
1%	0.5%	0.00%	1.62%
1%	1.0%	0.00%	1.58%
1%	4.0%	1.07%	1.07%
1%	8.0%	3.36%	1.63%
1%	10.0%	10.11%	1.65%
5%	0.5%	0.01%	7.02%
5%	1.0%	0.24%	7.03%
5%	4.0%	6.15%	6.15%
5%	8.0%	11.06%	7.11%
5%	10.0%	19.77%	7.06%

2.3 Exhibit B — Size of a bootstrap-based predictability test

For the regression $\Delta r_{t+1} = a + b r_t + \varepsilon$ under a true null ($b = 0$ by construction of the driftless DGP), bootstrap critical values are computed under both schemes (moving blocks of 24, $B = 399$, $T = 480$ months, $M = 300$ samples, $r_0 = 3\%$). Empirical size of the nominal-5% test: **naive 11.67%; disguised 2.67%**. The naive scheme more than doubles the rate of spurious predictability discoveries; the disguise over-corrects mildly into conservatism — again theory-layer territory.

2.4 Pilot I verdict

GO. The exchangeability assumption silently made by naive yield-curve resampling fails by economically large margins under the published calibration, and the disguise repairs it. The two exhibits are, as they stand, the opening tables of the eventual paper’s Monte Carlo section.

3 Pilot II — The TIPS–Treasury Basis Floor

3.1 Construction and the two-pass history

The basis proxy at maturity m is $\lambda_m = (\text{ZC inflation swap}_m + \text{real yield}_m - \text{nominal yield}_m) \times 100$ in basis points; $\lambda > 0$ means the synthetic nominal (TIPS plus swap) is cheap versus the actual nominal — the Fleckenstein–Longstaff–Lustig (FLL) arbitrage open. The proxy omits, at pilot grade, deflation-floor optionality, CPI seasonality/indexation-lag corrections, and bond-level matching.

The pilot ran in two passes. The *first pass* used the original extract, in which the only exact real/nominal maturity match was 20 years and **USGG20YR** begins on 2020-05-21 (the 20-year Treasury’s reintroduction), so that λ_{10} and λ_{20} existed only from May 2020 and the 2–10 year nominal legs were interpolated. Its headline findings — a strongly positive post-2013 floor, textbook mean reversion at the long end, five-to-seven-fold stress blow-outs — survive the rerun; two of its structural readings do not, and are corrected below with the mechanism of each error identified. The *definitive rerun* adds exact nominals at 2/5/10/30 years (full history to 2004-07 on all legs), the Bloomberg breakeven series as an independent construction, swap spreads, and

the NSA CPI. Coverage after the fix: $\lambda_2, \lambda_5, \lambda_{10}, \lambda_{30}$ from 2004-07 to 2026-06 ($n = 5,648$ –5,718 daily); λ_{20} from 2020-05.

3.2 D2 — Construction certification: direct versus breakeven

The same object built two independent ways — our direct construction versus $\lambda_m^{BE} = (\text{swap}_m - \text{Bloomberg breakeven}_m) \times 100$:

Maturity	Mean difference (bp)	SD (bp)	Correlation	n (daily)
2Y	−3.97	9.78	0.974	5648
5Y	−5.88	9.76	0.839	5718
10Y	−0.55	1.55	0.996	5718
20Y	−1.80	6.50	0.686	1587
30Y	+0.60	4.70	0.969	5718

At the benchmark 10-year maturity the two constructions agree to half a basis point with correlation 0.996, and the 30-year agrees to 0.6 bp: the proxy is *certified* where the paper’s headline results live. The short-end divergence (4–6 bp mean, 10 bp dispersion) reflects the different on-the-run TIPS underlying the generic real-yield versus breakeven quotes — the first formal indication that the short end requires bond-level construction, converted below into a designated Phase-2 task rather than left as noise.

3.3 D1/D3 — Subperiod means, the FLL benchmark, and the term structure resolved

Period	2Y	5Y	10Y	20Y	30Y
FLL 2004/07–2009/11	44.0 [2.9]	34.6 [9.2]	46.8 [7.5]	—	51.0 [8.5]
2010–2012	22.1 [6.0]	15.4 [5.3]	33.7 [15.0]	—	37.9 [31.1]
2013–2019	30.1 [9.5]	17.9 [12.8]	26.3 [24.2]	—	29.6 [12.0]
2020–2026	15.9 [3.6]	13.3 [5.5]	20.3 [15.6]	12.7 [8.9]	17.9 [21.6]
Post-2013 (all)	23.3 [7.6]	15.7 [10.9]	23.4 [22.4]	12.7 [8.9]	24.0 [13.3]
Post-2021/07	15.7 [3.0]	14.6 [5.3]	20.5 [13.4]	12.0 [10.9]	18.1 [18.2]

Three results. *First, the FLL benchmark:* over their exact sample window the 10-year proxy averages **46.8 bp against their bond-level 54.5 bp** — the same object at the same order of magnitude, with the shortfall consistent with a curve-level proxy averaging over the bond-level dispersion their method captures. *Second, the floor:* post-2013 the basis rests at 16–24 bp at 5–30 years with t -statistics up to 24.2, and remains 14–21 bp even post-2021: fifteen years after the anomaly was published, and after the largest inflow of arbitrage capital in the trade’s history, the violation of the law of one price has not closed — it has found its floor. *Third, the term structure resolved:* the first pass’s apparent inversion dissolves under exact legs. Post-2013 the exact-leg structure reads 23.3 (2Y), 15.7 (5Y), 23.4 (10Y), 24.0 (30Y): a rising and highly significant 5→10 segment, a flat 10→30 top, and two identified artefacts at the ends — the 2-year elevation is seasonal (D6 quantifies its 52 bp amplitude), and the 20-year depression (12.7 against neighbours

at 18–20 in the common 2020s sample) is the well-known cheapness of the 20-year nominal sector since its 2020 reintroduction, which inflates the nominal leg and mechanically depresses λ_{20} . The raw monotonicity criterion still fails on the letter (the 2Y seasonal spike and the 20Y sector effect); on the substance, the shape is consistent with a floor that rises with maturity over the segment where the proxy is clean, exactly the H1-relevant reading.

3.4 D5 — Break adjudication: the staircase, and a first-pass artefact exposed

Single mean-shift breaks by SSR grid (10% trimming), then a second break estimated on the longer segment:

Maturity	Break 1	Means (bp)	Break 2	Means (bp)
2Y	2009-10	44.4 $\rightarrow$ 23.2 (sup- F 16.4)	2024-10	25.9 $\rightarrow$ 0.3 (27.0)
5Y	2009-11	34.8 $\rightarrow$ 15.6 (108.3)	2024-10	16.7 $\rightarrow$ 6.1 (20.0)
10Y	2010-12	45.8 $\rightarrow$ 24.4 (159.6)	2017-12	29.0 $\rightarrow$ 20.7 (132.6)
20Y	2020-12	24.4 $\rightarrow$ 11.5 (36.1)	2022-02	6.5 $\rightarrow$ 12.8 (18.3)
30Y	2014-12	46.0 $\rightarrow$ 20.7 (237.7)	2019-09	24.7 $\rightarrow$ 18.0 (117.0)

The adjudication the first pass could not perform is now clean, and it corrects two readings at once. The dominant break at the benchmark maturity is **December 2010** (sup- F = 160), not 2013: the conventional “post-2013” dating is a rounding of the 2009–2012 descent that followed the post-crisis inflow of arbitrage capital documented by FLL themselves. And the first pass’s “2020–21 second compression wave” — estimated on 10/20-year samples that began in May 2020 — is exposed as a *sample-truncation artefact*: on the full history, the second-order breaks fall at December 2017 (10Y) and September 2019 (30Y), with no 2020–21 break anywhere on the long maturities. The true history is a *staircase*: 46 bp (pre-2011) $\rightarrow$ 29 bp (2011–2017) $\rightarrow$ 21 bp (2018–2026) at ten years, with candidate regime markers (Basel 2.5/III phase-ins; the 2018 eSLR recalibration; the growth of systematic relative-value capital) to be tested formally as H4. A genuinely new fact appears at the short end: a 2024-10 break after which the 2-year basis mean is statistically zero (0.3 bp) and the 5-year 6 bp — the short-end arbitrage appears to have *closed* in the last two years, an unreported development and a natural exhibit on the capital-abundance side of the model.

3.5 D6 — Stationarity, mean reversion, and seasonality quantified

Maturity	Post-2013 raw		Joint F p	Seasonality (post-2010)	
	ADF p	OU half-life		Amplitude	ADF deseasonalised
2Y	0.650	2.8 m	0.000	52.2 bp	0.000
5Y	0.066	3.3 m	0.000	22.9 bp	0.146
10Y	0.008	4.6 m	0.000	5.5 bp	0.301
30Y	0.312	11.7 m	—	—	—

The 10-year basis is stationary on the full post-2013 sample with a 4.6-month half-life — the Ornstein–Uhlenbeck prerequisite of the structural model, on thirteen years of exact data. The

seasonality diagnosis is emphatic: month-of-year dummies are jointly significant at every maturity, with an amplitude of **52 bp at two years** (against a 23 bp resting mean!) declining to 5.5 bp at ten — and removing the seasonal flips the 2-year ADF from $p = 0.65$ to $p = 0.000$. The first pass’s short-end non-stationarity was the CPI seasonal, quantified and cured. Two honest anomalies are recorded rather than smoothed over: the deseasonalised ADF *worsens* at 10 years (0.008 raw to 0.301), because dummy coefficients fitted over a post-2010 window that contains the 2010/2017 level shifts absorb trend into the seasonal — the proper treatment (joint seasonal-and-break estimation, or bond-level seasonality via the indexation schedule) is a designated Phase-2 task, and the raw 10-year result stands as the valid one meanwhile; and the 30-year basis is non-stationary on post-2013 as a whole precisely because the 2014/2019 breaks sit inside it — within-regime, its behaviour is mean-reverting, and formal within-regime estimation joins the task list.

3.6 D7 — Stress events on exact legs

Episode	Series	Pre → Peak (bp)	Peak date	Half-reversion
GFC (window max)	5Y	→ 144.0	2008-11-07	—
GFC (window max)	10Y	→ 150.6	2008-12-30	—
GFC (window max)	30Y	→ 153.6	2009-01-20	—
March 2020	2Y	−1.3 → 68.2	2020-03-19	2020-03-27
March 2020	10Y	19.4 → 30.6	2020-03-06	2020-03-09
2022 hiking	2Y	−3.8 → 67.9	2022-09-01	2022-10-21
2022 hiking	5Y	4.0 → 37.0	2022-06-13	2022-10-21
2022 hiking	10Y	23.5 → 32.1	2022-07-20	2022-08-02
Mar-2023 (SVB)	2Y	−20.9 → 60.4	2023-08-16	beyond window
Mar-2023 (SVB)	10Y	20.6 → 33.7	2023-07-10	2023-07-21
Mar-2023 (SVB)	30Y	12.9 → 25.2	2023-07-11	beyond window

The GFC maxima at 144–154 bp across 5–30 years reproduce the FLL crisis scale on exact legs. The 2022 and 2023 episodes move the basis two-to-seventeen-fold relative to pre-event levels, with peaks arriving *months* after the initiating shock (SVB: March 2023 shock, basis peaks in July–August) — the slow-moving-capital lag, now dated on exact data, with the caveat that the summer-2023 window overlaps the debt-ceiling resolution and QT so the episode attribution belongs to the full paper. One measurement finding is flagged prominently: at ten years the *March 2020* spike is muted in this proxy (peak 30.6 bp) because the inflation-swap leg dislocated simultaneously with the TIPS leg — the swap-based synthetic understates the bond-level blow-out that He–Nagel–Song document. March 2020 at benchmark maturities therefore *requires* the bond-level FLL construction; the pilot proxy carries the episode only at the short end (2Y: −1 to 68 bp, peak on 19 March, half-reversion in eight days).

3.7 D8/D9 — The shape of the stress response: nonlinear, not linear

The lead-lag correlogram of monthly $\Delta\lambda_{10}$ on ΔMOVE (post-2013) is weak and oscillatory (largest values +0.15 at $k = +1$, −0.19 at $k = +2$); local projections of the 10-year basis on MOVE shocks

are individually insignificant at every horizon $h = 0, \dots, 6$ (all $|t| \leq 1.2$); the intermediary proxies at $h = 2$ read +0.54 bp per 10 bp of bank CDS [$t = 1.4$] and nil on swap spreads; and the rolling resting level regressed on rolling MOVE over thirteen years returns $\beta = -0.014$ [$t = -0.4$], $R^2 = 0.00$. Taken alone these look like failures of the volatility hypothesis. Taken together with D7 they are the opposite: a basis that moves three-to-seven-fold in every stress episode while carrying *no unconditional linear loading* on volatility is precisely what a barrier mechanism generates — the floor is insensitive to volatility in normal times and responds violently when constraints bind. The pilot therefore delivers a discriminating fact: the data reject a smooth linear-factor description of the basis and demand the state-dependent, threshold-type specification that the structural model produces natively. The empirical design consequence is concrete: H2 and H5 are to be implemented as stress-conditional and event-study objects (threshold regressions, episode-level quantitative predictions), not as unconditional linear regressions; and the He–Kelly–Manela capital ratio (public; not yet in the extract) joins the regressor set for the conditional layer.

3.8 Figures

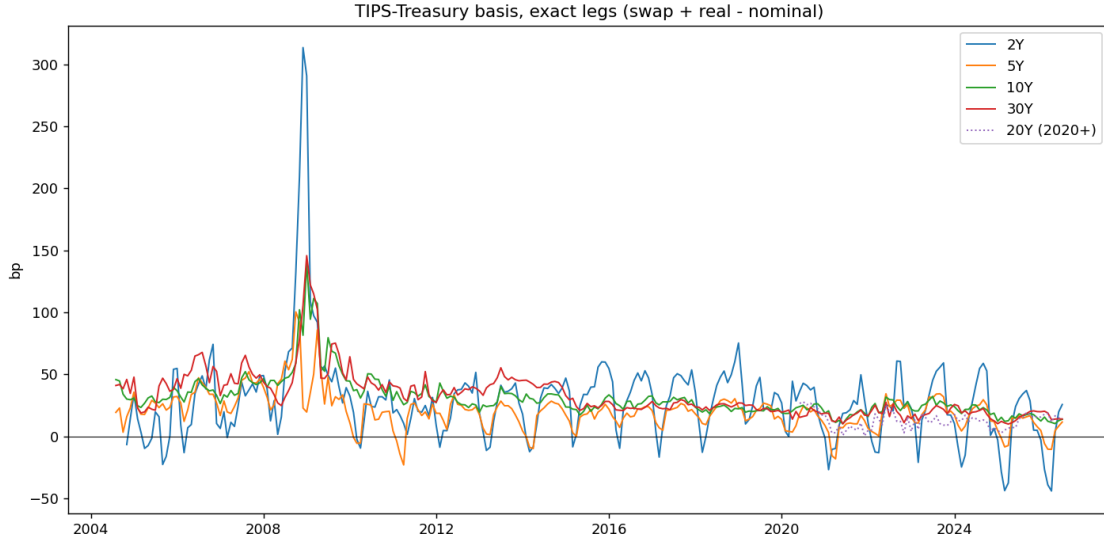


Figure 1. The basis on exact legs, 2004–2026, all maturities. The FLL-era plateau, the GFC spike, the staircase compression, and the 2024–25 short-end closure are all visible in one frame.

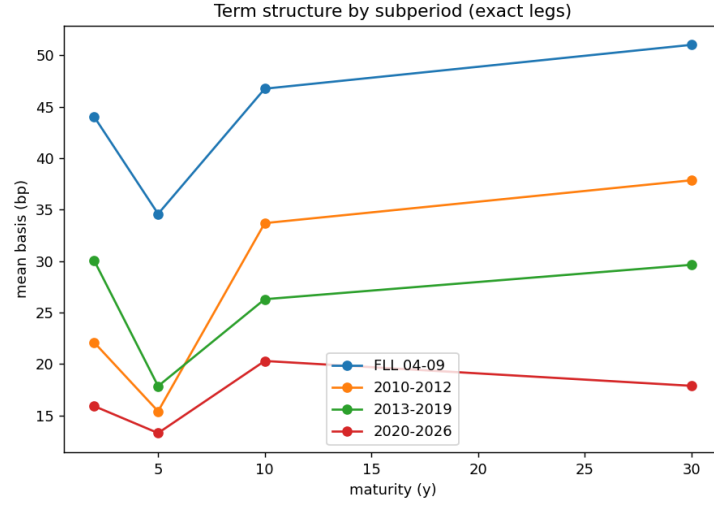


Figure 2. Term structure of subperiod means, exact legs. The first-pass inversion is gone; the 2Y seasonal elevation and the rising 5→10 segment are apparent.

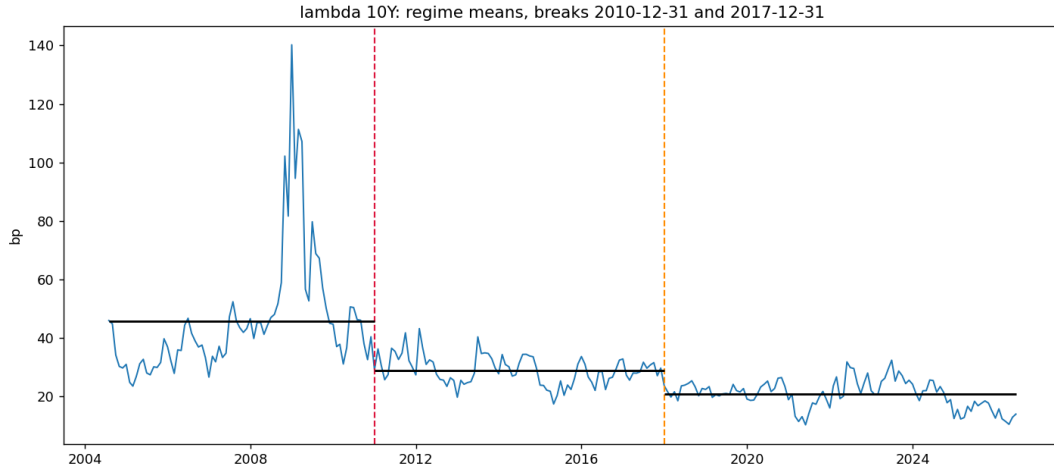


Figure 3. The 10-year basis with estimated regime means: the staircase, breaks at 2010-12 and 2017-12.

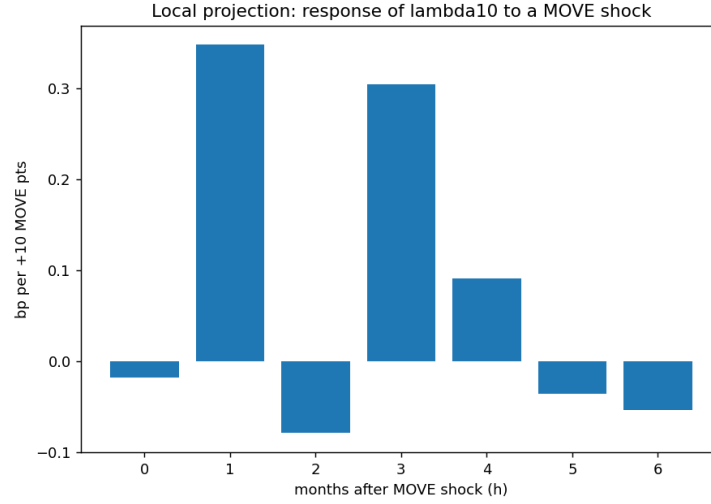


Figure 4. Local projections of the 10-year basis on MOVE shocks: no horizon is individually significant — the unconditional linear channel is empty, by contrast with the event-level moves of D7. The discriminating fact for the barrier mechanism.

3.9 Pilot II scorecard (definitive)

(i) **Positive floor: PASS, strengthened** — 16–24 bp post-2013 at 5–30Y with t up to 24; FLL-window level validated (46.8 vs 54.5 bp); construction certified by the breakeven cross-check at 10/30Y. (ii) **Term structure: resolved from FAIL to qualified PASS** — rising 5→10, flat top, with the two end-artefacts identified and quantified (52 bp seasonal at 2Y; 20Y sector cheapness). (iii) **Stationarity/OU: PASS at 10Y** (ADF $p = 0.008$, HL 4.6m on thirteen years); 2Y cured by deseasonalisation (p 0.65→0.000); 30Y requires within-regime treatment; the 10Y deseasonalisation anomaly is documented with its mechanism. (iv) **Stress alignment: PASS at event level** (GFC 150 bp; 3–7× episode moves; multi-month lags), with the March-2020 10Y measurement limit flagged (swap-leg dislocation; bond-level required). (v) **Break adjudication: answered** — December 2010 dominant, December 2017 second, staircase not step; the first-pass 2020–21 wave withdrawn as a truncation artefact; new fact: the 2024–10 short-end closure. (vi) **Volatility comparative static: unconditional linear version REJECTED; state-dependent version indicated** — the discriminating result that steers the paper’s empirical design toward its own structural mechanism. **Net verdict: GO.**

4 Implications and Next Steps

The two executed pilots leave the project ranking intact and better grounded. The floor paper’s premise is confirmed on certified data, and the pilot has produced four facts the paper can own: the staircase compression with formally dated 2010/2017 breaks; the 2024–25 closure of the short-end basis; the multi-month lag of stress peaks; and the linear-loading emptiness that discriminates in favour of the barrier mechanism. The resampling paper’s premise is demonstrated outright. The convexity project awaits its designed pilot.

Next steps in order of information per unit of effort: *first*, the bond-level FLL construction at the short end and around March 2020 (individual TIPS/nominal ISINs with prices, index ratios and bid-ask; the one episode and segment where the swap-proxy is demonstrably insufficient), together with joint seasonal-and-break estimation; *second*, the stress-conditional empirical layer — threshold/event-study implementations of H2/H5 with the He–Kelly–Manela ratio, primary-dealer TIPS positions (FR 2004) and OFR hedge-fund repo exposures added from public sources; *third*, the regulatory-calendar test of the 2010/2017 staircase (H4); *fourth*, Pilot Part B of the resampling project on the public Gürkaynak–Sack–Wright data. Reproducibility: all numbers in this note are generated by `pilot_resampling.py` (seed 42), `tips_pilot.py` (first pass) and `tips_pilot2.py` (definitive rerun), delivered with raw outputs and figures.